THE OPERATION OF CAPPING WITH A FIRST CHERN CLASS FACTORS THROUGH RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The definition of capping with a first Chern class and of rational equivalence. Same chapter chow.tex, labels definition-cap-c1 (proof line 4962) and definition-rational-equivalence (proof line 4963). Fixes the operation whose descent is being proved and rewrites the triviality of the cycle as a locally finite sum of principal divisors.
- The projection formula for capping with a first Chern class. Same chapter chow.tex, label lemma-pushforward-cap-c1, invoked at proof line 4973. Moves the cap through the proper morphism from the disjoint union of the components, reducing the statement to a single integral scheme.
- Commutativity on an integral scheme and additivity in the invertible sheaf. Same chapter chow.tex, labels lemma-commutativity-on-integral (proof line 4985) and lemma-c1-cap-additive (proof line 4990). The first rewrites the cap with the Chern class of L against the divisor of f as the cap with the trivial invertible sheaf against the divisor of a meromorphic section, and the second shows that this last expression vanishes.

Source reference. Original definition 02SO: sources/stacks-chow/pinned-original-source.tex, source lines 4284–4284. Navigation pointer for \ref{definition-cap-c1} used inside the counted span (line 4962); the argument that consumes it is retained in the span.

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Source reference. Original definition 02RW: sources/stacks-chow/pinned-original-source.tex, source lines 2882–2882. Navigation pointer for \ref{definition-rational-equivalence} used inside the counted span (line 4963); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SP: sources/stacks-chow/pinned-original-source.tex, source lines 4330–4330. Navigation pointer for \ref{lemma-c1-cap-additive} used inside the counted span (line 4990); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TH: sources/stacks-chow/pinned-original-source.tex, source lines 4924–4924. Navigation pointer for \ref{lemma-commutativity-on-integral} used inside the counted span (line 4985); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SU: sources/stacks-chow/pinned-original-source.tex, source lines 4631–4631. Navigation pointer for \ref{lemma-pushforward-cap-c1} used inside the counted span (line 4973); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4949); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$ be invertible on X. The operation $\alpha \mapsto c_1(\mathcal{L}) \cap \alpha$ factors through rational equivalence to give an operation*

$$c_1(\mathcal{L}) \cap - : \mathrm{CH}_{k+1}(X) \to \mathrm{CH}_k(X)$$

Proof. Let $\alpha \in Z_{k+1}(X)$, and $\alpha \sim_{rat} 0$. We have to show that $c_1(\mathcal{L}) \cap \alpha$ as defined in Definition 02SO is zero. By Definition 02RW there exists a locally finite family $\{W_j\}$ of integral closed subschemes with $\dim_\delta(W_j) = k + 2$ and rational functions $f_j \in R(W_j)^*$ such that

$$\alpha = \sum (i_j)_* \mathrm{div}_{W_j}(f_j)$$

Note that $p : \coprod W_j \to X$ is a proper morphism, and hence $\alpha = p_*\alpha'$ where $\alpha' \in Z_{k+1}(\coprod W_j)$ is the sum of the principal divisors $\mathrm{div}_{W_j}(f_j)$. By Lemma 02SU we have $c_1(\mathcal{L}) \cap \alpha = p_*(c_1(p^*\mathcal{L}) \cap \alpha')$. Hence it suffices to show that each $c_1(\mathcal{L}|_{W_j}) \cap \mathrm{div}_{W_j}(f_j)$ is zero. In other words we may assume that X is integral and $\alpha = \mathrm{div}_X(f)$ for some $f \in R(X)^*$.

Assume X is integral and $\alpha = \mathrm{div}_X(f)$ for some $f \in R(X)^*$. We can think of f as a regular meromorphic section of the invertible sheaf $\mathcal{N} = \mathcal{O}_X$. Choose a meromorphic section s of $\mathcal{L}$ and denote $\beta = \mathrm{div}_{\mathcal{L}}(s)$. By Lemma 02TH we conclude that

$$c_1(\mathcal{L}) \cap \alpha = c_1(\mathcal{O}_X) \cap \beta.$$

However, by Lemma 02SP we see that the right hand side is zero in $\mathrm{CH}_k(X)$ as desired. □